DDLS 2025
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Project plan

Background

Nuclear Magnetic Resonance (NMR) spectroscopy is an analytical technique that exploits the magnetic properties of atomic nuclei to study molecular structure. When placed in a strong magnetic field, nuclei with spin (such as ^{1}H, ^{13}C) can absorb radiofrequency energy and resonate at characteristic frequencies. ^{1}H-NMR (proton NMR) is the most common type, since nearly all organic molecules contain hydrogen. Each unique chemical environment around a hydrogen atom causes a slightly different resonance frequency (chemical shift), producing peaks in the spectrum. Peak splitting (multiplets) reveals information about neighboring atoms, while intensities reflect the number of protons, enabling detailed structural interpretation.

Scientific question

Can we use supervised machine learning to classify small molecules into broad metabolite families (e.g., amino acids, sugars, lipids, nucleotides) based on their ^{1}H-NMR peaklist spectra?

Scientific relevance

Metabolomics relies on NMR for rapid, non-destructive profiling of metabolites in biological samples. Assigning metabolites to broad chemical families from spectral data is useful for:

- Early biomarker discovery in precision medicine.
- Automated metabolite identification pipelines.
- Reducing manual spectral interpretation workload.

Dataset

Main dataset

Source: Human Metabolome Database (HMDB, hmdb.ca/downloads)

License and terms of use:

> *HMDB is offered to the public as a freely available resource. Use and re-distribution of the data, in whole or in part, for commercial purposes requires explicit permission of the authors and explicit acknowledgment of the source material (HMDB) and the original publication (see below). We ask that users who download significant portions of the database cite the HMDB paper in any resulting publications.*

Files:

- Raw NMR Spectra Peaklist Files (TXT) - ppm/intensity peaklists for many metabolites.
- All Metabolites Metadata (XML) - compound metadata, structures (SMILES, InChI), and classifications (superclass, class, subclass).

Linking: TXT filenames contain HMDB IDs (e.g., HMDB0000161) that map directly to entries in the XML metadata.

Subset: 500-1000 metabolites covering at least 4-5 broad families (Amino acids, Sugars, Lipids, Nucleotides, Organic acids).

Supplemental datasets (if needed)

Biological Magnetic Resonance Bank (BMRB)

Exploration Goals

Before modeling, we will explore:

- Spectral structure: How many peaks per compound? Distribution of ppm ranges across classes?
- Class balance: Are families equally represented, or will we need class weighting / balancing?
- Spectral similarity: Do spectra from the same class cluster in PCA space?
- Technical variation: Are there solvent/field strength differences that need normalization?

Proposed Model & Evaluation

Baseline Model

- Random Forest classifier on binned spectra (e.g., 0.02 ppm buckets).
- Logistic Regression as a second simple baseline.

Improved Models

- Support Vector Machine with RBF kernel.
- 1D Convolutional Neural Network (CNN) treating spectra as a sequence.
- Autoencoder for feature reduction, then classification.

Evaluation

- Metrics: Accuracy, weighted F1-score.
- Good performance definition: >70% accuracy across classes, better than random guessing and naive baselines.
- Cross-validation: 5-fold CV to ensure robustness.

Accessibility

Wrapped pipeline in web application, which lets users upload NMR spectra to determine the metabolite group.

Feasibility

Expected runtime:

- Preprocessing (binning, metadata join): <5 minutes.
- Model training (Random Forest, SVM): seconds–minutes on CPU.
- CNN training: <30 minutes on Colab GPU, still feasible on CPU with reduced dataset.

Risks:

- Large XML metadata file (≈6-9 GB) → handled by streaming parser, only extracting needed IDs.
- Class imbalance (e.g., lipids more abundant than amino acids) → mitigated by balanced sampling or weighting.
- Spectral variability (instrument/solvent effects) → normalized by scaling and binning.

Overall: Highly feasible in 2-3 weeks.

ChatGPT brainstorm

Link to ChatGPT brainstorming session.

Appendix

ChatGPT “Deep Research” report

Deep Research Report — ¹H-NMR Peak-List Classification

Project: DDLS 2025 Final Project — Alfred Larsson
Date: 2025-10-08
Agent: ChatGPT (GPT-5 Thinking)

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1) Objective

Run a "Deep Research" to survey existing research, alternative approaches, and relevant data sources for classifying small molecules into metabolite families from **¹H-NMR peak-list spectra**. Provide concrete sources, data access points, and methodological options suitable for a 2–3 week project.

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2) Key Findings at a Glance

- **Data sources:** HMDB (peak-list TXT, spectrum pages), BMRB (metabolomics entries, GISSMO library), nmrshiftdb2 (open database of assigned shifts & prediction), and COLMAR (query, ppm tools, HSQC databases).
- **Methods used in literature:** Random Forest, SVM/PLS-DA/OPLS-DA baselines; CNNs and autoencoders for 1D spectra; newer **GNNs/transformers** for shift prediction and spectral learning.
- **Augmentation:** **Predict chemical shifts** from structures (COLMARppm, PROSPRE, nmrshiftdb2) to expand training data and **normalize by solvent/field**.
- **Feasibility:** HMDB + BMRB provide enough compounds to build a multi-class classifier across superclasses (Amino acids, Sugars, Lipids, Nucleotides, Organic acids) within the course constraints.

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3) Datasets & Access Points

3.1 Human Metabolome Database (HMDB)

- **Peak-list bundle:** "**Raw NMR Spectra Peaklist Files (TXT)**" (contains ppm/intensity lists).
- **Per-spectrum pages** provide *List of chemical shift values (TXT)* and **nmrML/JDX**.
- **License/terms:** Free to use with acknowledgement; commercial use requires permission.
- **Notes for this project:** HMDB TXT files carry HMDB Accessions; match to XML metadata for **Class/Superclass** labels.

Links:

- Downloads index (shows "Raw NMR Spectra Peaklist Files (TXT)").

- Example spectrum pages with **TXT peak lists** in water at 500 MHz.

3.2 BMRB (Biological Magnetic Resonance Data Bank)

- **Metabolomics portal** with bulk downloads (entries tarball, relational CSV/SQL).
- **GISSMO** library provides curated small-molecule spectra; downloadable in **NMR-STAR** and **NMReDATA** formats.
- **Use in project:** Supplement HMDB with additional compounds; harmonize to **peak lists** via vendor tools or nmrML parsers if needed.

3.3 nmrshiftdb2

- Open database supporting **assigned shifts** and **prediction** (^{1}H/^{13}C).
- **Use in project:** Generate **predicted ^{1}H peak lists** (solvent/field aware if available) to **augment** minority classes or to pretrain an **autoencoder**.

3.4 COLMAR / COLMARppm

- Web tools and databases (incl. HSQC for mixtures) and **COLMARppm** for **rapid ^{1}H/^{13}C shift prediction** and automated assignment.
- **Use in project:** Benchmark or augment with predicted shifts; check ppm tolerances and solvent conditions.

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4) Methods Landscape & Alternative Approaches

4.1 Supervised Baselines (Shallow)

- **RF / SVM / Logistic / PLS-DA** on **binned ^{1}H spectra** (e.g., 0.01–0.04 ppm bins).
- **Pros:** Fast, interpretable feature importances, aligns with your current pipeline.
- **Refs:** Evaluation studies comparing multivariate classifiers on simulated/experimental ^{1}H-NMR; themed reviews on ML in NMR.

4.2 Deep Learning for 1D ^{1}H Spectra

- **1D-CNN** for sequence-like spectra; **denoising autoencoders** → classifier.
- **Spectrum-aware models:** e.g., **NMRQNet** for identification/quantification from NMR spectra.
- **Transformers / contrastive**: pretrain on predicted/real spectra (SimCLR-style) to learn ppm-invariant embeddings.

4.3 Graph-Based Chemical-Shift Predictors (for Augmentation)

- **GNN-based** chemical-shift prediction from small data; **PROSPRE** (deep learning, **solvent-aware** ^{1}H shift prediction).
- **Augmentation idea:** From **SMILES** → predicted **^{1}H shifts** → synthesize peak lists (add line-shapes if needed), then train classifier; helpful for **class imbalance**.

4.4 Domain-Specific Tools

- **COLMARppm** for fast shift predictions and assignments (can generate clean synthetic peaks).
- **GISSMO** curated lineshapes aid realistic synthetic spectra (Lorentzian/Gaussian).

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5) Practical Recommendations for This Project

1. **Core dataset:** HMDB peak-list TXT + matched **Superclass/Class** from HMDB XML (stream parse).
2. **Supplement:** Add BMRB/GISSMO entries where classes are under-represented.
3. **Preprocessing:**
   - **Solvent/field normalization:** keep water/D2O and 500 MHz subsets; otherwise apply **ppm alignment** and **intensity scaling** (unit max or total integral).
   - **Binning:** 0.02 ppm bins in [-1, 14] ppm is a good start; test 0.01 and 0.04.
   - **Peak to vector:** convert lists to binned intensities; optionally **peak-picking → sparse vector**.
4. **Baselines:** RF, Logistic, and **PLS-DA** with 5-fold, report **balanced accuracy & weighted F1**.
5. **Improvements:**
   - **1D-CNN** on normalized vectors; small architecture (100k params).
   - **Autoencoder pretraining** or **contrastive pretraining** (real+predicted spectra).
   - **Class weights** or **SMOTE-like augmentation** with **predicted shifts** for minority classes.
6. **Error analysis:** Confusion matrix by **Superclass**; inspect ppm regions driving misclassifications (use permutation/SHAP on bins).
7. **Accessibility:** Minimal **Streamlit** app: upload peak list → binned vector → prediction + top-bins explanation.

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6) Risks & Mitigations

- **Instrument/solvent heterogeneity:** Restrict to consistent conditions (e.g., H\ O/D\ O, 500 MHz) or apply alignment; document solvent/field in features.
- **Class imbalance:** Use **class weights** and **shift-prediction augmentation** (COLMARppm/PROSPRE).
- **XML size (HMDB): Streaming parse** to extract only accession + class fields.

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7) Citations & Source Pointers

- HMDB **Downloads** (lists “Raw NMR Spectra Peaklist Files (TXT)”): *Human Metabolome Database — Downloads.*

- Example HMDB spectrum pages with **TXT lists** and nmrML/JDX: examples for 500 MHz H\ O.
- **BMRB metabolomics** bulk downloads and relational DB; **GISSMO** library page.
- **nmrshiftdb2** overview & usage; GNN shift-prediction on nmrshiftdb2 data (Rull 2023).
- **COLMARppm** (Rigel 2023/2024) — ACS Anal. Chem. article & PubMed; free PMC article.
- **NMRQNet** (Wang 2023) for deep NMR analysis.
- **Evaluation of multivariate classifiers** for 1D ^{1}H-NMR (Vu 2019).
- **Tea origin classification** via ^{1}H-NMR + ML (Hou 2024) — an applied example.
- Review: **Machine learning in NMR spectroscopy** (Klukowski 2025).
- Perspective on metabolomics & COLMAR databases (Markley 2017).

(Full URLs and metadata embedded in the accompanying transcript below.)

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8) Search Transcript (Prompts & Results)

Prompt 1: "HMDB NMR peaklist files proton NMR peaks TXT download HMDB"
Result: HMDB downloads page lists "Raw NMR Spectra Peaklist Files (TXT)".

Prompt 2: "classification of metabolites from 1H NMR spectra machine learning study"
Results: NMRQNet (2023); classifier comparisons for 1D ^{1}H-NMR (2019); applied studies (e.g., 2024 tea origin).

Prompt 3: "BMRB small molecule NMR database metabolomics peak lists download"
Results: BMRB metabolomics portal; bulk downloads; GISSMO library.

Prompt 4: "nmrshiftdb2 dataset download predicted 1H NMR chemical shifts small molecules"
Results: nmrshiftdb2 site, documentation; shift-prediction GNN paper (2023).

Prompt 5: "COLMAR NMR database small molecules 1H 13C metabolomics"
Results: COLMARppm articles (2023/2024) describing fast ^{1}H/^{13}C shift prediction and assignment.

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9) How to Cite in Your Plan

- Cite HMDB downloads and specific spectrum pages you used (include accession IDs).
- If using BMRB/GISSMO, cite the metabolomics portal and GISSMO.

- If augmenting with predicted shifts, cite **COLMARppm** and/or **PROSPRE** with solvent details.

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10) Appendix: Direct Links

- HMDB Downloads — "Raw NMR Spectra Peaklist Files (TXT)".
- HMDB per-spectrum examples (TXT lists, 500 MHz H\ O).
- BMRB metabolomics portal; bulk downloads.
- GISSMO library.
- nmrshiftdb2 home; usage guide; Rull 2023 GNN shift prediction.
- COLMARppm (ACS Anal. Chem. paper; PubMed; PMC open text).
- NMRQNet (PMC).
- Classifier evaluation for 1D ^{1}H-NMR (PMC).
- Machine learning in NMR spectroscopy (2025 review).
- Metabolomics perspective incl. COLMAR (2017).
- Applied example (tea origin classification, 2024).

Prepared for inclusion as the "Deep Research transcript/report" in the DDLS Step-1 Project Plan.